

The Anatomy of “Beating PPO”: Exploration Walls, the Value Pathway, and Where Model-Based Advantages Invert

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Abstract

Claims that model-based or structured methods “beat PPO” rarely survive matched-environment controls. We run TD-MPC2, tuned brax PPO, and tuned brax SAC on byte-identical MJX environments and subject every headline claim to an adversarial control. Three findings survive. (1) **A categorical on-policy exploration wall exists but is narrow:** on HopperHop, tuned PPO never reaches return 200 in 5/5 seeds at 472M steps, while SAC crosses by ~ 8 M and TD-MPC2 by ~ 1 M; on the same robot’s easier Stand task the wall is graded (PPO escapes 2/8; SAC solves 5/6 at 5M; TD-MPC2 3/3 by 0.9M), and five control tasks show no wall at all. At a matched 1M budget, TD-MPC2 solves both hopper tasks while neither model-free method solves either. (2) **The mechanism is value learning, not dynamics modeling:** in a five-loss ablation replicated at $n \geq 4$ on three tasks, removing the TD value loss or the policy prior individually reproduces total failure—including the exploration wall itself ($n=6$: the hop is never found)—while the reward head matters only for planning and the self-predictive consistency loss is always the mildest cut. (3) **The advantage ordering is task-class-dependent and inverts on a 21-DoF humanoid:** TD-MPC2’s training diverges to NaN on 7/7 default-config runs (and its only numerically stable run never learns), PPO is unusable under three configurations, and untuned SAC solves 4/5 — on the high-DoF morphology only plain off-policy learning works. We conclude that “world model beats PPO” decomposes into a sample-efficiency and consistency advantage carried by the TD value pathway, real but confined to a characterizable dynamics class, with no evidence of a level advantage anywhere.

1 Introduction

Model-based reinforcement learning methods such as TD-MPC2 [5] report large advantages over on-policy baselines like PPO [9]. Such comparisons are sensitive to environment mismatches, configuration errors, and seed-starved rate estimates. We present a case study in adversarial self-review: every headline claim we initially formed was rewritten by its own control experiment, and we report the final, control-surviving form together with the audit trail.

Our contributions:

- A matched-environment three-arm benchmark protocol (TD-MPC2 / tuned PPO / tuned SAC on byte-identical MJX tasks), with two upstream configuration bugs found and documented by the audit (a task-name case mismatch silently skipping tuned PPO hyperparameters; a missing humanoid configuration causing NaN under defaults).
- A budget-indexed characterization of the on-policy exploration wall (Section 3).
- A five-loss ablation of TD-MPC2 replicated on three tasks at $n \geq 4$ per arm, isolating the value pathway as individually necessary (Section 4).
- A boundary result on a 21-DoF humanoid where the ordering inverts (Section 6).

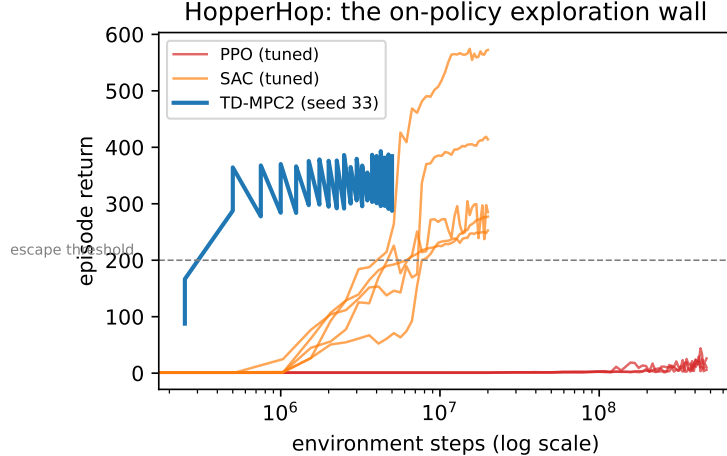


Figure 1: HopperHop seed curves on byte-identical MJX environments (log- x): tuned PPO stays below the escape threshold through hundreds of millions of steps; tuned SAC crosses at 4–8M; TD-MPC2 crosses within 1M.

2 Setup

All methods interact with identical `mujoco_playground` MJX environments (`registry.load(env, impl=jax)`, episode length 1000, action repeat 1, verified byte-identical across arms). PPO and SAC use the library’s tuned configurations (2048 / 128 parallel environments); TD-MPC2 uses SimNorm latents with MPPI planning (2048 samples, horizon 3). Returns are best evaluation returns unless stated. *Config audit lesson*: the tuned-parameter lookup silently returns defaults on any name mismatch; we audit resolved configurations for every run after discovering that `PendulumSwingUp` vs. `PendulumSwingup` voided the tuned override and manufactured a spurious “wall.”

3 The exploration wall, budget-indexed

HopperHop — categorical for PPO. Tuned PPO: peak 53.8, 0/5 seeds ≥ 200 through 472M steps/seed; the wall also survives the standard exploration knob (entropy cost $\times 3$ and $\times 10$: peaks 4–74, $n=2$ each at 150M). On the *graded* HopperStand barrier, by contrast, entropy $\times 3$ put 1/4 seeds over the threshold (627 vs. 118–150; $\times 10$: 0/4, 116–185) — a rate consistent with the baseline’s 2/16 escape lottery rather than a repair. The knob neither repairs the graded barrier nor dents the categorical wall. SAC crosses 200 in 6/12 seeds by 5M and 6/9 by 8M (the 20M cohort crossed 5/5 at 4.1–7.7M; four direct-8M seeds went 1/4 — the crossing by 8M is a $\sim 2/3$ rate, not a certainty). TD-MPC2: 6/6 by ~ 1 M; 5M bests 295–610, mean 420 ± 113 (sd, $n=12$).

HopperStand — graded but near-categorical. PPO escapes 2/16 (eight further seeds at the full 285M budget all walled at 105–195; both escapes, 681/749, peaked within 120M). SAC: 0/3 at 1M; 7/10 by 5M (solves 464–932; failures 33/100/300). TD-MPC2: 3/3 by ≤ 0.9 M (962/948/943; the third seed by 0.3M). The final three-method gradient — TD-MPC2 3/3 $\gg$ SAC 7/10 $\gg$ PPO 2/16 — and the matched-1M column (*TD-MPC2 solves both hopper tasks at a budget where neither model-free method solves either*) are the headline statements.

No-wall controls. FingerTurnHard (PPO ≈ 973 , 3/3), Pendulum (solved once the config bug was fixed: 842/831), BallInCupCatch (discovery-luck for both algorithms), CheetahRun (PPO 892–922 at 285M, matching SAC 918/912 at 10M; an earlier “never ≥ 500 ” scoring was a budget/config artifact we retract), WalkerRun. The wall is therefore specific to the single-leg hopper’s unstable, contact-timing-critical dynamics — categorical only for the hop itself.

Discriminator: AcrobotSwingup (unstable, contact-free). PPO learns normally (267–344, $n=4$ at

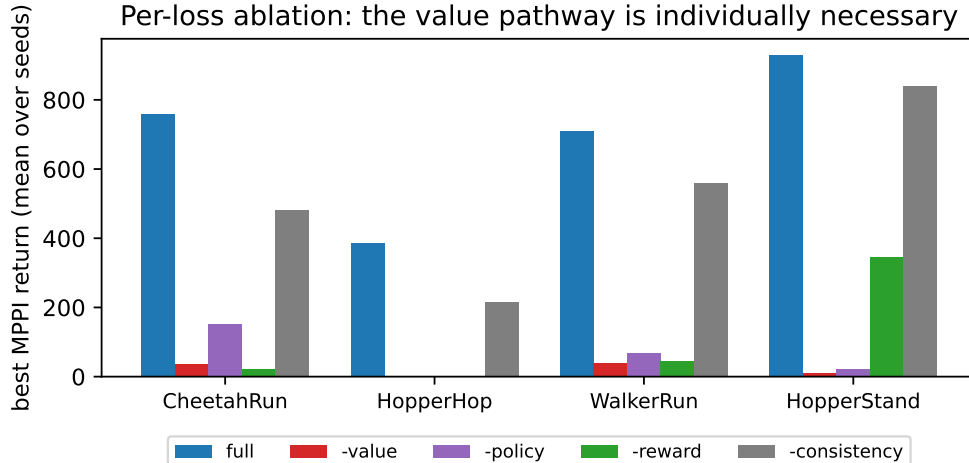


Figure 2: Five-loss ablation across four tasks (best MPPI return, mean over seeds, $n \geq 2$ per bar; decisive arms $n \geq 4$). Removing the TD value loss or the policy prior is fatal everywhere; removing the self-predictive consistency loss barely registers.

285M) — *the wall requires contact-criticality, not instability alone*. Conversely SAC is slow and inconsistent here even at 20M (42/207; 54–89 at 5M): the off-policy advantage is also task-class-specific. TD-MPC2 leads on both classes (429/422 within 1M) — the value-pathway advantage is the most portable of the three. (TD-MPC2 HopperStand later firmed to 5/5: 962/948/943/959/959.)

4 Which loss does the work? The value pathway

We zero one loss term at a time (from scratch; masks verified in logs) and evaluate both MPPI and policy-only returns, on CheetahRun, HopperHop, and WalkerRun, $n \geq 4$ per decisive arm (HopperHop value cell $n=6$).

Arm	CheetahRun (MPPI best)	HopperHop	WalkerRun
full	738/782/721/795	287–570 (6/6 find gait)	699–723
–value	16/37/58 (dead)	0.0/0.0/3.2/0.0/0.0/0.1 ($n=6$) — gait never found	38–39 (dead)
–policy	MPPI limps 123–192; π dead ≤ 11	≈ 0 both readouts	53–83 (dead)
–reward	MPPI 5–31; π 761–805 (full)	MPPI ≈ 0 ; π 189–519	MPPI 44; π 681–723
–consistency	367–558 (π to 623)	$\sim$ half strength, 4/4 find gait	483–674

Table 1: Five-loss ablation. Removing the TD value loss reproduces the exploration wall itself; the self-predictive dynamics loss is the only component whose removal merely degrades.

The pattern replicates on a fourth task, HopperStand ($n=4$ /arm), which the full model solves in 0.3M steps: none 911–946 (all solve); value-ablated 6–13 and policy-ablated 9–34 — dead; reward-ablated MPPI 265–542 but π 943–944; consistency-ablated 816–898 — nearly no cut at all. *Without the TD value loss, TD-MPC2 cannot even learn to stand.*

The ablation’s mildest cut is not uniformly removable, and the pattern is sharper than a task-class split. Training consistency-OFF from scratch at the full 5M budget: on HopperHop the stripped agent scores 165/475/481/511 ($n=4$) against the full model’s 420 ± 113 ($n=12$) — three seeds at the top of the full band, reaching 440+ by ~ 2 M — but on every other task tested the stripped agent loses badly: WalkerRun 537–594 (mean 565) vs. 709–782 (mean 737), -23% ; CheetahRun 516–528 (mean 524) vs. 782–904 (mean 849), -38% ; AcrobotSwingup 233–352 (mean 284) vs. 488–533 (mean 511), -44% — all non-overlapping ($n=4$ per cell). Acrobot is exploration-flavored, so “removable when exploration-bound” does not survive;

the reading that fits all four tasks is that the self-predictive loss underwrites MPPI rollout quality wherever the *planner* carries learning, and HopperHop — where the policy head learns the behavior directly — is the one task where it is redundant.

“The world model explores” thus decomposes: the TD value signal trained through the latent discovers and ranks behavior; the self-predictive dynamics loss is a helpful regularizer, not the key.

5 Level vs. efficiency; hierarchy control

At $4\times$ TD-MPC2’s budget (20M), SAC finals on HopperHop are 246/277/285/414/572 ($n=5$) — the distributions overlap TD-MPC2’s $\sim 412\text{--}481$. The world model’s robust advantages are *consistency* and $\sim 4\text{--}5\times$ sample-efficiency, not a level ceiling. A dense-shaping control also re-attributes a feudal-hierarchy “win” (4/6 vs. flat 0/6) to signal density: shaped flat TD3 solves 3/6; the surviving claim is that the hierarchy *self-generates* that signal (replicating 7).

6 The boundary: a 21-DoF humanoid inverts the ordering

On DMC HumanoidWalk: PPO produces `reward=nan` under all three configurations tried (no official tuned humanoid configuration exists upstream). TD-MPC2’s training loss diverges to NaN on 6/6 default-config walk seeds (onsets 0.53–2.51M) and on the easier Stand (0.28M); best return before any divergence was 30.4. Reducing the update-to-data ratio shifts the onset stochastically ($k=32$: NaN at 3.13M; $k=64$: the only run of nine to finish 4M) — and the surviving run never learns (best 21.8). Untuned SAC solves 4/5 (625–909 at 5M; 1/5 NaN). Config/numerical robustness is thus a failure axis orthogonal to exploration, and any “X beats PPO” claim must state which axis it lives on.

7 Related work

Sample-efficiency victories for model-based and hybrid methods are well replicated (EfficientZero [10], DreamerV3 [4], TD-MPC2 [5], RLPD [1]); asymptotic-level victories keep eroding as model-free baselines are scaled and regularized (BRO [8], SimBa [6], TD7 [3], CrossQ [2]); most measurable HRL benefit reproduces on flat agents given better exploration/shaping [7]. Our results agree from an independent, matched-environment direction and add the per-loss attribution and the high-DoF inversion.

8 Discussion

Every one of our initial headline claims (“gait-specific wall,” “morphology-categorical wall,” “world model buys level,” “hierarchy beats flat”) died under one more seed or one more control, always toward a smaller, truer statement. The surviving picture: structured priors and world models buy speed and consistency, never ceiling; their engine is value learning; and their advantage region is a characterizable dynamics class outside of which the ordering can fully invert. We offer the audit trail (public ledger and blog) as a template for adversarial self-review.

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